

MASTER ORGANIZATIONAL RULE FOR THE AUDIO PRODUCTION & ENGINEERING GLOSSARY, QUESTION BANK, AND MOBILE APP

Governing framework for all future glossary terms, definitions, flashcards, quiz questions, diagrams, study modules, and mobile app content.

1. Core Principle

All glossary terms and concepts shall be organized primarily by the Audio Production & Engineering program course structure rather than solely by technical discipline.

Terms may appear in multiple courses when appropriate. The same term should be introduced at an introductory level in early courses and revisited in later courses with progressively greater technical depth, practical application, and diagnostic complexity.

This curriculum follows a spiral learning model in which concepts are scaffolded from basic recognition and operation to advanced analysis, design, optimization, troubleshooting, and professional practice.

2. Primary Source Corpus

The master glossary is based on five core reference documents:

- Audio Production Glossary
- Audio Production Glossary - Volume 2: Industry-Specific Terms
- Microphone Components and Audio Tube Glossary
- Audio Terms and Definitions
- Audio Measurement and Smaart Glossary

Together these documents provide the foundational vocabulary for recording, acoustics, electronics, live sound, measurement, diagnostics, broadcast, game audio, AV integration, and related professional fields.

3. Program Course Organization

- MUSI 190 - Introduction to Music Technology
- MUSI 201 - Recording Arts
- MUSI 202 - Computer Music and MIDI Production
- AUDI 201 - Sound Reinforcement Systems
- AUDI 204 - Audio System Design and Maintenance
- MUSI 205A - Audio Production Projects I (Audio-Only Media)
- MUSI 205B - Audio Production Projects II (Visual and Interactive Media)

4. Course Definitions and Subject Coverage

MUSI 190 - Introduction to Music Technology

The foundational survey course of the entire program. Students learn the basic tools, terminology, and operational concepts used throughout audio production and engineering.

Topics include:

- Fundamentals of sound and acoustics
- Human hearing and psychoacoustics
- Pitch, frequency, amplitude, loudness, and decibels
- Basic electronics concepts
- Signal flow and gain structure
- Connectors and cabling
- Microphone types and polar patterns
- Proximity effect, pads, and high-pass filters
- Audio interfaces and preamps
- Mixing consoles and channel strips
- Equalizers and filters
- Compressors, gates, and effects processors
- Digital audio fundamentals
- Basic Digital Audio Workstation operation
- Recording terminology
- Loudspeaker and amplifier fundamentals

Depth level: Introductory. Broad coverage with practical recognition and basic operation, but without extensive technical detail.

Depth range: Levels 1-2

MUSI 201 - Recording Arts

A focused study of professional recording techniques and studio practice.

Topics include:

- Advanced microphone theory
- Microphone placement and distance
- Stereo recording techniques
- Instrument-specific microphone methods
- Signal path from microphone to preamp to converter to DAW
- Studio acoustics and room interaction
- Recording workflow and session management
- Mixing techniques
- Editing
- Mastering fundamentals

Depth level: Intermediate to advanced. Greater technical and artistic depth with extensive practical application.

Depth range: Levels 2-4

MUSI 202 - Computer Music and MIDI Production

Focused on composition and production using computers and MIDI.

Topics include:

- MIDI protocol and sequencing
- Sampling
- Synthesis
- Drum programming
- Loop construction
- Virtual instruments
- Software instruments
- Automation
- Arrangement
- AI-assisted music-production tools

Depth level: Intermediate to advanced with emphasis on computer-based creation and composition.

Depth range: Levels 2-4

AUDI 201 - Sound Reinforcement Systems

Focused on live sound, installed sound, and applied audio systems.

Topics include:

- Sound reinforcement principles
- Analog and digital mixing
- Loudspeakers and amplifiers
- Monitor systems and FOH
- Feedback control
- Assisted listening systems
- Commercial audio and 70V/100V distribution
- Corporate AV systems
- Vehicle audio
- Dante networking
- Audio-over-IP
- System deployment and operation

Depth level: Intermediate to advanced with emphasis on real-world deployment and system operation.

Depth range: Levels 2-4

AUDI 204 - Audio System Design and Maintenance

Focused on design, installation, repair, and maintenance of audio systems and components.

Topics include:

- Audio system design principles
- Basic electronics
- Analog circuit theory
- Vacuum tube electronics
- Microphone components
- Soldering
- Cable fabrication and repair
- Connectors and wiring
- Equipment cleaning and maintenance
- Troubleshooting and diagnostics
- Preventive maintenance

Depth level: Advanced with strong emphasis on design, component-level understanding, and repair.

Depth range: Levels 3-5

MUSI 205A - Audio Production Projects I (Audio-Only Media)

Focused on professional audio projects that do not require synchronized visual media.

Topics include:

- Podcast production
- Voice-over recording
- Narrative audio
- Radio production
- Spoken-word editing
- Audio storytelling
- Band recording
- Mixing and mastering for audio-only delivery

Depth level: Advanced project-based application of recording and production skills.

Depth range: Levels 3-5

MUSI 205B - Audio Production Projects II (Visual and Interactive Media)

Focused on audio for picture and interactive media.

Topics include:

- Film audio
- Video post-production
- ADR
- Foley
- Sound design
- Game audio
- Interactive audio
- Multimedia production
- Advanced band recording

Depth level: Advanced to professional application integrating audio with visual and interactive media.

Depth range: Levels 3-6

5. Program-Wide Advanced Topics

These topics may appear in multiple courses and are treated as cross-disciplinary subject areas:

- Architectural acoustics
- Measurement and diagnostics
- Rational Acoustics Smaart
- System optimization
- Broadcast and streaming
- Commercial AV integration
- Home theater
- Standards and professional organizations

6. Term Tagging Rules

- Term
- Definition
- Primary technical discipline
- Applicable course or courses
- Depth level within each course
- Related terms
- Practical examples
- Quiz and flashcard associations
- Source references

7. Depth Level Framework

Level	Meaning
1	Recognition and basic terminology
2	Operational understanding and simple use
3	Applied technical understanding

Level	Meaning
4	Analysis and optimization
5	Design, maintenance, and repair
6	Professional and expert mastery

8. Course Depth Mapping

Course	Depth Levels
MUSI 190	Levels 1-2
MUSI 201	Levels 2-4
MUSI 202	Levels 2-4
AUDI 201	Levels 2-4
AUDI 204	Levels 3-5
MUSI 205A	Levels 3-5
MUSI 205B	Levels 3-6

9. Implementation Rule

Whenever new glossary terms, questions, diagrams, or study materials are created:

- Determine all courses to which the term applies.
- Assign a depth level for each course.
- Preserve overlap where educationally appropriate.
- Present introductory explanations in earlier courses.
- Expand to deeper technical and professional explanations in later courses.
- Do not reorganize this course structure unless explicitly directed to do so.

10. Guiding Principle

The Audio Production & Engineering glossary is a curriculum-mapped knowledge system. The course structure is the primary organizational framework, while technical disciplines function as secondary tags. Concepts are intentionally repeated across courses with increasing sophistication so students build mastery through progressive exposure and application.